

David Wu

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INTRODUCTION

Junior

NTU

Taiwan, Taipei

Sep. 2023 - Now

- I am a junior student studying Computer Science in National Taiwan University. I am interested in TCS and algorithms, as well as following the latest development in cryptology and quantum field. I am familiar with C and Git, and have practical experience on algorithm implementation, design data structures and low-level system programming. Given my research experience and lectures, I can make knowledge practical, discuss, and improve with my team, also give contribution to the project and the company.

EDUCATION

National Taiwan University

Bachelor of Computer Science and Information Engineering, Minor in Atmospheric Sciences

Taiwan, Taipei

Sep. 2023 - Now

EXPERIENCE

Fast Crypto Lab

Academia Sinica

July 2025 - Present

Taipei, Taiwan

- Working on the project of Big Number Multiplication, which aims to find the crossover point between GMP's flexible algorithms and our NTT implementation on Arm A-profile CPUs.
- Employing engineering procedure such as unit tests and benchmarking on our implementation, also doing pipelining and optimization using hand-written assembly without compiler.

PROJECTS

Operating Systems Projects | C

Spring 2025

- Set up and extended xv6 in a Docker-based environment for modular testing and development across multiple machine problems
- Designed and implemented a user-level thread library with cooperative scheduling and context switching primitives
- Built a SLAB memory allocator with support for freelist management and partial/full slab classification.
- Developed real-time and non-real-time schedulers (HRRN, P-RR, DM, EDF-CBS) and simulated RAID-1 fault-tolerant file operations.

Fuzzy Counting with Probabilistic Data Structures | C++, Python

Spring 2025

- Designed and implemented a typo-tolerant fuzzy matching framework using Bloom filter segmentation and Cuckoo filter for fuzzy lookup.
- Prototyped HyperLogLog and HyperLogLog-LDP matchers in Python, supporting q-gram segmentation and similarity-based voting.
- Conducted experiments across varying (m, q, st, k) parameters to analyze trade-offs in accuracy and space under synthetic typo datasets.
- Build a Python library for autocorrection use, which is already uploaded.

CAPTCHA Generation and Evaluation Platform | Python

Spring 2025

- Developed a modular CAPTCHA defense platform integrating image generation, adversarial perturbations, and model-based attack evaluation.
- Implemented visual noise types (Gaussian, Salt-and-Pepper, affine transforms, JPEG compression) and evaluated their impact on ML-based solvers.
- Built a Streamlit-based GUI for real-time CAPTCHA creation, model inference, and visual feedback on attack results.

COURSES

A+, A: Computer Programming, Linear Algebra, Machine Learning, Cryptography and Network Security, Operating Systems, Advanced Data Structure

TECHNICAL SKILLS

Languages: Java, Python, C/C++, HTML/CSS, Arm Neon Intrinsics, Assembly

Developer Tools: Git, Docker, VS Code